

A Reflection of My General Education Courses at Iowa State University

When I first asked what engineers do, the answer I got was that they solve problems to make the world a better place. That's true, but it leaves something out. A solution only works if people can understand it, afford it, and be willing to use it. My technical classes are teaching me how to build things, but my general education classes taught me to think about the people those things are built for. They didn't make me better at coding or math. Instead, they changed the questions I ask before I start.

The biggest change was learning to question my own assumptions. In INTST 2500, we moved from region to region, studying climates, economies, and cultures that were completely different from what I grew up with. RUS 3750 did something similar by following Russia from Gorbachev to Putin. I was fascinated by how quickly the Soviet Union fell apart, and I was struck that people in Russia today often can't protest their government because they risk going to jail. I take speaking up for granted, and that class showed me that what feels normal to me isn't normal everywhere. HDFS 2760: Human Sexuality made the same point from another angle. We studied how culture and gender expectations shape the way people think and make choices, and I was surprised how often people treat their own norms as universal. For an engineer, this matters. If a team assumes every user thinks like they do, they'll build something that works well for some people and leaves out the rest. I now try to gather more than one perspective before settling on a solution, even though it takes more time up front.

Some of the most memorable lessons showed me that the "obvious" solution isn't always the right one. In RUS 3750, the topic that stuck with me most was Chernobyl. After the accident, people were forced to leave their homes because the radiation was too dangerous. Yet some of the older generation still lives inside the contamination zone because they refused to go. They feel at peace in the place they've always lived. From a purely technical view, moving everyone out is the safe answer, but for those people, home meant more than safety. INTST 2500 gave me a similar lesson with ivory smuggling in Africa. I assumed it was a simple poaching problem, but governments there run entire task forces to stop it, and I was shocked when I learned that officials had found a whole shipping container full of tusks. The problem involves local law enforcement, international shipping, and a lot of money. Both examples taught me that real problems are rarely as simple as they first appear, and that a solution has to account for the people, politics, and economics around it.

Another skill I gained is working within limits. In HDFS 2830: Personal Finance, we learned calculations like debt-to-income ratios, future value, and interest rates, but I learned the most from the case studies, where we had to apply those ideas to situations that could really happen to someone. Working through them helped me feel more prepared for my own life, and it changed how I think about cost. The price tag is only part of the story. Something cheap up front can cost far more over time in repairs and replacements, and if the people who need a solution can't afford it, it doesn't matter how well it was engineered. Almost every engineering project has a budget, a deadline, or a regulation to work around, and I'm more comfortable now making decisions inside those limits instead of getting stuck on what I can't change.

These lessons connect directly to problems society faces today. Global supply chains are under strain, and a disruption in one region can delay products and raise prices everywhere else. After studying how geography, politics, and resources shape different regions in INTST 2500 and RUS 3750, I understand why a design that depends on a single supplier or country is risky. The rising cost of living has made affordability a social issue as well, and Personal Finance taught me to look at lifetime cost and consider what people with limited incomes can realistically afford. Sustainability is related, since products that are built to last or be repaired usually cost less over time and create less waste. As someone who will likely work in software, I'll also have to think about the digital divide and data privacy. A website that assumes fast internet or a new device leaves out rural and lower-income communities, and the systems I build will often hold personal information that people trust me to protect. The cultural awareness from HDFS 2760 helps here, because it taught me to listen to people and ask questions instead of guessing what they need. None of these problems can be solved with technical skill alone.

Looking back, none of these classes taught me how to build a webpage or structure a database, but they taught me to ask better questions before I begin. Who is this for? What can they afford? What do they believe? Who might I be leaving out? And what is life like where they live? As I start my career, I want to be an engineer who takes those questions as seriously as the technical ones, because that is what turns a working solution into one that actually improves the world around us.